

Sections: B,C & D

Linear Algebra (MT-1004)**Homework#29****Instructor: Muhammad Ali**Name: **Muneeba Zaheer**Roll No: **23i-0581**Section: **CS-30****Regression using QR Factorization****Question 1**Find the QR factorization of the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 8 \\ 1 & 7 & 13 \\ 1 & 9 & 21 \end{bmatrix}$$

Calculate R by using two different method.**method 1:-**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 8 \\ 1 & 7 & 13 \\ 1 & 9 & 21 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $u_1 \quad u_2 \quad u_3$

using dot product method to make orthogonal:

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = u_2 - \text{proj}_{v_1} u_2$$

$$= \begin{bmatrix} 2 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix} - \frac{\begin{bmatrix} 2 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

dividing by magnitude $\frac{2\sqrt{205}}{5}$

$$= \begin{bmatrix} 2 \\ 3 \\ 5 \\ 7 \\ 9 \end{bmatrix} - \frac{26}{5} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -16/5 \\ -11/5 \\ -1/5 \\ 9/5 \\ 19/5 \end{bmatrix} = \begin{bmatrix} -8/\sqrt{205} \\ -11/\sqrt{205} \\ -1/\sqrt{205} \\ 9/\sqrt{205} \\ 19/\sqrt{205} \end{bmatrix}$$

$$v_3 = u_3 - \text{proj}_{v_1} u_3 - \text{proj}_{v_2} u_3$$

$$v_3 = \left(\begin{bmatrix} 3 \\ 6 \\ 8 \\ 13 \\ 21 \end{bmatrix} - \begin{bmatrix} 3 \\ 6 \\ 8 \\ 13 \\ 21 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right) - \left(\begin{bmatrix} 3 \\ 6 \\ 8 \\ 13 \\ 21 \end{bmatrix} - \begin{bmatrix} 3 \\ 6 \\ 8 \\ 13 \\ 21 \end{bmatrix} \begin{bmatrix} -8/\sqrt{205} \\ -11/2\sqrt{205} \\ -1/2\sqrt{205} \\ 9/2\sqrt{205} \\ 19/2\sqrt{205} \end{bmatrix} \right)$$

$$v_3 = \begin{bmatrix} 37/41 \\ 71/164 \\ -247/164 \\ -237/164 \\ 265/164 \end{bmatrix}$$

$$\frac{v_3}{\|v_3\|} = \begin{bmatrix} 0.90 \\ 0.43 \\ -1.51 \\ -1.45 \\ -1.62 \end{bmatrix}$$

$$\|v_3\| = 1.93$$

$$Q = \begin{bmatrix} 0.4 & -0.5 & 0.3 \\ 0.4 & -0.3 & 0.1 \\ 0.4 & -0.03 & -0.5 \\ 0.4 & -0.3 & -0.5 \\ 0.4 & -0.6 & 0.5 \end{bmatrix}$$

$$R = Q^T A = \begin{pmatrix} 2.25 & 11.7 & 22.35 \\ 0 & 5.73 & 14.1 \\ 0 & 0 & 2.86 \end{pmatrix}$$

Method 2:-

using online calculator to compute values:-

$$v_1 = u_1$$

$$v_2 = u_2 - \text{proj}_{v_1} u_2$$

$$u_2 = v_2 + \text{proj}_{v_1} u_2$$

$$v_3 = u_3 - \text{proj}_{v_1} u_3 - \text{proj}_{v_2} u_3$$

$$v_3 = v_3 + \text{proj}_{v_1} u_3 + \text{proj}_{v_2} u_3$$

$$u_1 = v_1 + 0v_2 + 0v_3$$

$$u_2 = \frac{u_2 v_1}{v_1 \cdot v_1} v_1 + v_2$$

$$u_3 = \frac{u_3 v_1}{v_1 \cdot v_1} v_1 + \frac{u_3 v_2}{v_2 \cdot v_2} v_2 + v_3$$

$$A = \begin{bmatrix} v_1 & v_2 & v_3 \\ |v_1| & |v_2| & |v_3| \end{bmatrix} \begin{bmatrix} 1 & |v_1| & \frac{u_2 v_1}{v_1 \cdot v_1} \cancel{|v_1|} & \frac{u_3 v_1}{v_1 \cdot v_1} \cancel{|v_1|} \\ 0 & 1 & |v_2| & \frac{u_3 v_2}{v_2 \cdot v_2} \cancel{|v_2|} \\ 0 & 0 & 1 & |v_3| \end{bmatrix}$$

$$R = \begin{bmatrix} 2.24 & 11.62 & 22.36 \\ 0 & 5.73 & 14.1 \\ 0 & 0 & 2.85 \end{bmatrix}$$